TEND ROBOTICS

Keeping elderly people safe in their own homes.

Pre-read for Raj

We link in-home physical AI to a real nurse. The goal: one nurse watching one hundred homes.

Summary

Right now, somewhere, an older person has fallen at home and no one knows yet. Most older people want to stay in their own homes, but there aren't enough care workers to keep them safe there, and the number of workers is falling. The danger is the hours between visits, when no one is watching.

Devices that spot a fall and call for help already exist. Medical alert buttons and home sensors do this today for about $30-50 a month. But they can only send help. They can't judge what kind of help you need.

Tend adds a nurse. Sensors watch the home and flag when something's wrong. The nurse reads the alert and decides what to do: settle a false alarm, send a carer, or call an ambulance. The aim is one nurse covering a hundred homes. The nurse layer is the hard part, and it's the part other companies skip. That's what would make Tend hard to copy.

I chose this problem deliberately, using a [framework] from Hardy: a painful, worsening, structural problem. I am watching my grandparents live the consequences right now. My mum will not move into aged care. I worry every time I leave her at home, she has constant heart arrhythmia. I am building the thing my own family will need. I have a robotics background, and know people in health agencies.

I'm asking for $250,000 to test whether this works before I spend ten years building it. In three months I'll run the five tests that could prove the idea wrong. I'll talk to 300 people: carers, care companies, funders, and families. I'll put bought sensors into three to five real homes, sit in the alert stream myself for a few weeks, and measure the actual alert noise. Anything clinical escalates to a contracted nurse line. A yes means three things: customers willing to pay, alerts rare enough that one nurse can handle a hundred homes, and a legal path for remote triage, or onshore economics that still clear margin.

The problem

Somewhere tonight, an older person is on the floor of their own home, and no one knows yet. That gap, the hours between visits when nobody is watching, is the problem Tend exists to close.

Older people want to stay home, and the system cannot keep them there safely.

~80% of Australian older people want to remain in their current home.[1]

The workforce to honour that wish does not exist.

Over 59% of US home care agencies report ongoing caregiver shortages.[2] HCAOA survey data puts the average agency's unstaffable work at 510 hours a month.[3]

CEDA projects Australia will be short 110,000 direct-care workers by 2030, rising past 400,000 by 2050.[4]

New Zealand's aged residential care holds the worst nursing vacancy rate in the health system: 15% in 2023.[5]

The cost of care is crushing, and families pick up the slack.

Residential care runs roughly NZD 60-90k a year in New Zealand,[6] AUD 70-100k in Australia[7] and USD 115-130k in the US.[8] Means-tested subsidies shift part of this onto the state; the rest comes out of family estates. Every year an older person stays safely at home defers that entire bill. In the US alone, AARP estimates 59 million family caregivers provided 49.5 billion hours of unpaid care in 2024, worth just over USD 1 trillion.[9]

Why physical AI will close this gap

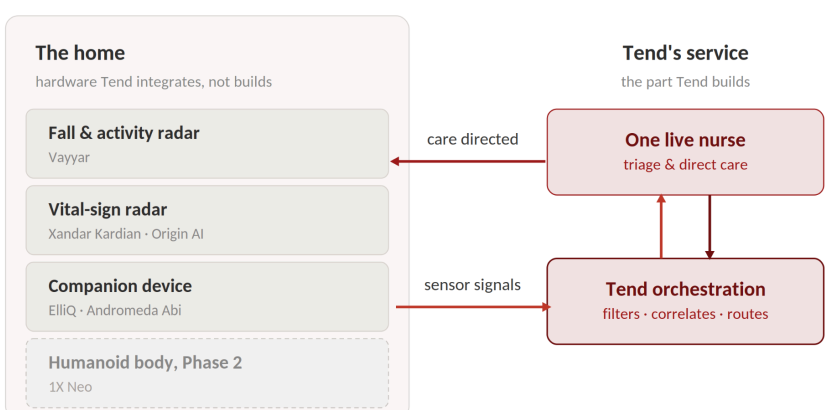
The technology to find that person on the floor of their home already exists. Contactless radar spots falls in care homes today, and monitored alarm services have run for decades. Companion robots, medication reminders and voice check-ins are well covered too. But none of these tools answers the real shortage: skilled people who decide what happens next. That decision layer, one step above detection, is the gap Tend closes.

The open position is clinician-grade triage. No one wires ambient detection to a licensed clinician who can triage, assess and direct care in seconds, in private homes, on a path to one nurse per hundred homes. Newly FDA-cleared radar devices make contactless vital-sign monitoring possible with nothing touching the person.[12]

Remote clinical labour changes the economics. An offshore or centralised nurse acts as a triage and escalation specialist. Existing carers keep delivering hands-on visits; the nurse decides which alerts matter and what happens next.

Orchestration across a smart home. It's reasonable to expect household humanoids to be deployed at scale in the next five years, and IoT devices are already commonplace. Tend will be the coordination layer that makes embodied AI deployable in a home care setting.

The moat is clinical and regulatory. The vetted clinician network, clinical governance, liability cover and payer contracts turn detection into a live clinical response.



The obvious objection, answered

The first question this idea earns: fall detection wired to a live responder has existed for decades at USD 30-50 a month, so what does Tend own?

Four things:

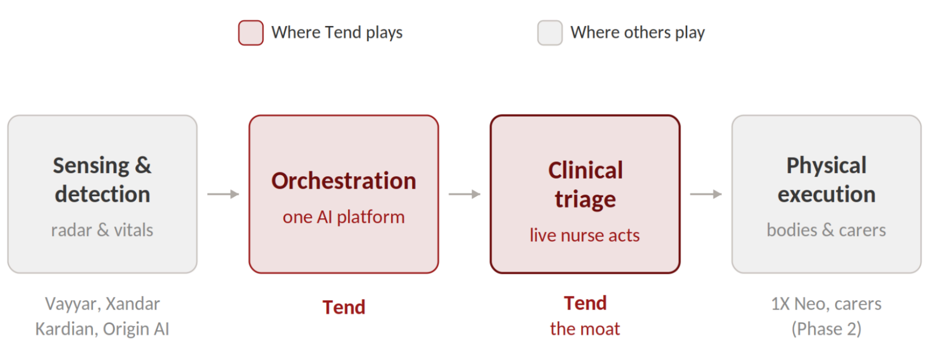
- **The escalation stack.** Tend builds the tooling that puts a live clinician on an alert within seconds. The nurse assesses, resolves most alerts without an ambulance, and directs care on the rest. We do not replace clinicians or carers. We make the ones already working far more productive.
- **The private home.** Incumbents deploy radar into residential care homes and telecare estates. Tend deploys into private homes, where competition is thinner. The funder may be the same, Te Whatu Ora or an insurer. The setting, the install, and the care model are not.
- **The clinical governance layer.** Tend builds what detection vendors avoid: triage protocols, liability cover and payer contracts. That layer works whether we contract existing telehealth nurses or employ our own.
- **The orchestration layer.** Tend runs one AI platform across every sensor and, later, humanoid bodies. No incumbent has (yet!) shown ambition toward mobility or teleoperation.

Two parts of this answer are hypotheses, and I treat them as such. Whether buyers pay a clinician premium over dispatch-grade pricing is tested in the paid pilots (Appendix 2). Whether a remote nurse can lawfully triage across borders is an open licensing question: whether a protocol-bound triage role can sit below the registration line (Appendix 3), with an onshore fallback costed in Appendix 2.

The vision

Phase 1: sensing plus companion plus a live clinician

We orchestrate proven components rather than build them. Buying the sensing layer puts our whole effort into the layer no vendor touches: the clinical network and the live response. Existing carers keep visiting, now only when required. Our system watches the gaps and can alert a clinician live.



Phase 1 capabilities:

- Detect falls, including someone found on the ground between carer visits.
- Prompt medication at the right time and confirm it was taken, where memory is unreliable.
- Check that the person is eating and moving through their day.
- Track breathing rate, heart rate, and movement through contactless radar. No wearable, nothing touching the person.
- Watch daily patterns like room transitions, bathroom use, and time out of bed, and flag changes.
- Surface risk signals to a live clinician, who decides whether and how to act.

Candidate components: Vayyar for falls, presence and activity; Xandar Kardian or Origin AI for vital-sign radar; ElliQ or Andromeda's Abi for the companion layer. A evaluation on the sensing layer is a month-one Frontier task (Appendix 1).

Phase 2: mobility and one nurse for one hundred homes

Once families trust a Tend device in the home, we add mobility and non-contact physical tasks.

We move into embodied robotics only once value is proven and the brand is trusted. We wait for bodies like Neo to be ready. These capabilities stay non-contact by design for now, which keeps us clear of medical-device regulation (Appendix 3).

Phase 2 capabilities:

- Move to the person and the incident. Bathrooms are a major risk area.
- Carry the nurse to the person for telepresence through teleoperation.
- Guide and lead: kitchen to a meal, door to a visitor, bed at night.
- Fetch and carry non-clinical objects around the home.
- Visually inspect the environment, then prompt and confirm at close range.

The destination is one remote nurse caring for one hundred homes.

As the technology keeps advancing, a robot will one day shower a person and cook them a meal. The tech is almost there. People are not ready yet, but trust is built slowly, and Tend will be the one they trust to do it.

Teleoperation is the bridge to autonomy. Early Phase 2 tasks run with a human in the loop, exactly as Neo ships today. When an incident happens, a nurse can take control through teleoperation and be present in seconds. As models improve, more of each task runs on its own, and one operator supervises more homes. The nurse-to-homes ratio that drives margin is a function of alert precision and task autonomy. Appendix 2 shows the arithmetic.

The Frontier plan

Months 1 and 2: understand deeply

- Run deep discovery with carers, providers, funders and families to pin down the exact moments where a live clinician creates value, and the ones where it does not. Target 300 conversations across the three months.
- Run evaluation on the sensing layer vendors, with Vayyar, Xandar Kardian and Origin AI as candidates. Test data access, unit costs, and any exclusivities.
- Convert existing relationships across government health agencies, home-care providers and the healthcare community into pilot and evidence partners.
- Commit fully. Move to Tend full-time, with the IP held cleanly in the company and my full attention on the build.

Month 3: test demand

- Build a scrappy MVP and sign at least five firms as paid design partners, with pilot pricing agreed. Aged-care procurement will not close full contracts in a month; signed paid intent is the honest test of pull.
- Run pilot triage by hand to learn the alert load, with a contracted 24/7 telehealth nurse line behind every clinical escalation. No roster gets built yet.

What would kill this

Frontier exists to kill weak ideas. These are the tests, with thresholds.

- **Nobody pays the premium.** If, after 300 conversations, no provider or funder will sign paid pilot intent at or above the placeholder band in Appendix 2, the clinician premium over dispatch-grade services fails.
- **Alerts are too noisy.** If field data from candidate sensors implies more than roughly ten clinician-minutes per home per day, one nurse per hundred homes is arithmetically out of reach, and the margin story fails.
- **Licensing blocks the ratio everywhere.** If counsel confirms remote triage into the home requires local registration in every target market, and onshore staffing cannot clear roughly one nurse per eighty homes at New Zealand rates, the long-run margin engine fails.
- **Homes refuse the sensing.** If a meaningful share of elders and families in discovery reject always-on sensing on privacy or dignity grounds, in-home deployment fails as designed.
- **No vendor opens the door.** If neither Vayyar nor an equivalent grants platform-level data access on workable terms, orchestration fails, and building radar hardware is the wrong company to build.

Any one of these ends the idea in its current form. I would rather learn that in month two than in year three.

Why Me?

My unfair advantage is domain fluency. I have spent my career advising large public sector organisations on AI. I understand how they evaluate technology, how funding decisions get made, and what evidence earns trust. Most founders spend their first year learning that.

My current work sits in the hardest part of AI: choosing what deserves to exist. I have run strategy and ideation across hundreds of AI use cases, cut the weak ones early, and carried the strong ones through to delivery. I have supported the stand up of AI and high-tech solutions in big enterprise and public sector environments repeatedly. That full cycle, from raw idea to a system people rely on, is exactly what Tend demands.

I have the technical foundation this problem demands. I trained in mechatronics and now lead Physical AI at Deloitte New Zealand, including our NVIDIA alliance. I understand the sensing and robotics stack from first principles. I know the ecosystem players before day one.

I build. I shipped BornNutrition in London and Tallow Lab alongside full-time work. My honours thesis became a working product rather than a report. Building while employed is hard, and I have done it repeatedly.

People trust me with access, and access is distribution in health. I hold quarterly catchups with Deloitte's CEO and was named APAC Rising Star and WIICTA Rising Star. Those relationships convert to pilot partners and evidence partners in month one.

I chose this problem deliberately. It is painful, worsening, and structural, and my advantages compound here. It can hold my attention for a decade.

The honest reason sits closer to home. My mum will not move into aged care when the time comes. I worry every time I leave her at home. She is ill. She randomly has dizzy spells and needs to put herself to bed. She is independent but not always. I am building the thing my own family needs.

Appendix 1: Competitors and components

The market splits into monitored response services, companion robots, monitoring software and home humanoids. None assembles the loop Tend assembles, though several own pieces of it.

Vayyar Care is the most direct challenge to our Phase 1 sensing claims, and our strongest candidate component. Vayyar ships contactless 4D imaging radar for fall detection today, at scale. Wall-mounted units detect falls automatically and monitor presence, time in bed, bathroom visits and activity patterns, without cameras or wearables.[11] Its commercial weight sits in residential care homes and telecare deployments rather than private homes. Alerts route to whoever the deployment defines: on-site staff in senior living, or 24/7 monitored response centres in telecare. Responders are trained call handlers. Vayyar does not employ nurses, hold clinical governance, or carry care liability.

Critically, Vayyar already partners. It runs an established partner programme and supplies radar to third-party telecare and monitoring platforms, so orchestrating its sensors asks nothing new of it. Open questions we test in month one: what data access a platform partner receives (processed alerts versus raw point cloud), unit and subscription costs at our scale, and whether exclusivities constrain a New Zealand or Australian deployment. The threat cuts the same way: a Vayyar monitoring partner could add clinician triage and close our gap. Speed into the clinical layer matters.

Monitored response is the incumbent category. US medical alert firms such as Bay Alarm Medical and Medical Guardian pair sensors with 24/7 professional monitoring. SafelyYou pairs fall detection with remote review in memory care. UK telecare routes radar detection to response centres within seconds.[11] This category proves demand for detection-to-response and sets the consumer price anchor near USD 30-50 a month. Its responders dispatch; they cannot clinically assess.

Intuition Robotics (ElliQ). A tabletop companion offering reminders, fall-risk assessment and a check-engine-light alert to clinicians. It has raised about USD 85 million.[13] Washington authorised a statewide Medicaid reimbursement code for ElliQ in March 2026, the first social AI robot fully reimbursed by Medicaid, which proves the US payer channel for in-home care devices.[14] It is expanding internationally, partnering with Kanematsu for a localised Japan launch in 2026.[15] It is not mobile and does not support teleoperation. It flags a human rather than connecting one who responds live. Two cautions: ElliQ has its own clinician-alert ambitions and owns that Medicaid channel, so platform access is a question to test rather than assume. And for our sensing list, Vayyar is arguably the better-fitting component.

1X (Neo) is the leading in-home humanoid. It ships in 2026, teleoperated first, with elderly assistance named as a target use case. It brings the body without the care loop.[16] We use this hardware if it stays ahead and switch makers if it does not. Staying body-agnostic keeps that choice ours.

Zuowei (Shenzhen Zuowei Technology) shows where China is going. It builds smart toileting and bathing robots and a broader eldercare platform, claiming China's first systematic range across toileting, bathing, eating, transfers, walking and dressing.[17] This is the contact-care frontier we deliberately avoid for now. It marks where the market heads once trust and regulation catch up.

Andromeda Robotics (Abi) is a companion, and a possible partner. It places friendly companion robots in aged care, with no monitoring or teleoperation and heavy EU biometric exposure. We have a relationship with Andromeda and may partner to use their operating system if needed.

Sensi.AI monitors remotely, but audio only. It detects and alerts. It does not connect a clinician who intervenes.

CareClever (Cutii) is the cautionary tale. A capable mobile companion sold B2C at 90 euro a month. It failed on consumer subscription economics. Our read, which is inference rather than record, is that carrying detection promises without a response service compounded the problem. Either way the lesson holds: sell the response, and sell it to whoever funds care.

The gap, stated precisely: companionship, monitoring and dispatch are covered ground. No one wires ambient detection to a licensed clinician who can triage, assess and direct care in seconds, in private homes, on a path to one nurse per hundred homes. On current evidence, that position is open.

Appendix 2: Business model hypothesis and first-pass costing

Everything here is first-pass arithmetic. Validated costing is a core Frontier deliverable, built from the 300 conversations and real vendor terms.

Who pays?

- Providers, such as home care agencies looking to increase revenue by increasing the number of patients per employed nurse.
- Funders (e.g. Te Whatu Ora contracts, Australia's Support at Home, US Medicaid HCBS and Medicare Advantage). They buy to mitigate the strain on the health system caused by those aging in place (keeping them out of hospital where possible) and to subsidise families' access to this technology, mitigating the economic pressure aging in place can put on families. ElliQ's statewide Washington Medicaid code shows the funder channel is real for in-home devices.[14]
- In New Zealand, how far this channel reaches depends on service classification. Personal care is funded at every income, while household help needs a Community Services Card. No category yet covers ambient monitoring with clinician response, so which test applies is a discovery question.

Price anchors. Dispatch-grade PERS sits near USD 30-50 a month. Residential care in New Zealand costs the equivalent of NZD 5,000-7,500 a month.[6] A placeholder band for discovery: NZD 600-900 per home per month for sensing plus live clinical response. This number exists to be attacked in discovery, and it will move.

What a clinician seat costs. A response measured in seconds needs an always-staffed seat. Round-the-clock coverage takes roughly 4.5 nurse FTEs per seat once leave and sickness are covered. At a loaded New Zealand registered nurse cost of about NZD 140k,[24] one seat costs about NZD 630k a year. Clinician cost per home per month, before hardware and overhead:

Nurse-to-homes ratio	Clinician cost per home per month
1:40 (onshore, early)	~NZD 1,310
1:100 (target)	~NZD 525
1:250 (autonomy-assisted)	~NZD 210

What sets the ratio. Homes per seat equals productive clinician minutes divided by clinician minutes per home per day. One seat holds 1,440 minutes a day; assume half is usable after surge headroom. At two minutes per home per day, one seat covers several hundred homes. At fifteen minutes, fewer than fifty. False-alarm rates from field data decide which world we are in, which is why alert precision is kill criterion two.

The onshore fallback. If offshore triage is barred (Appendix 3), the model must clear margin onshore. At the placeholder price band, onshore economics need the ratio above roughly one nurse per eighty homes. That is the sensitivity the licensing work must resolve.

Pilot staffing within the SAFE. The $250k SAFE does not fund a 24/7 nurse roster. For pilots we contract escalation capacity from an existing 24/7 telehealth nurse line, and we price pilots to cover that marginal triage cost. The roster gets built once the ratio maths and the funding channel are proven.

Appendix 3: Regulatory position

The claim this rests on

This pre-read says Tend is not a medical device. That claim is load-bearing, so it needs precision rather than absolutes.

Regulators do not classify a whole company. They classify each function by what it does and what it claims to do. Tend has two functions that sit on opposite sides of the line.

The sensing layer can stay outside device regulation. Contactless monitoring of activity, sleep and movement, framed as wellness and untied to a named disease, falls under the low-risk general wellness policy.[18]

The escalation function is the sensitive one. The moment software issues a clinical alert on its own, it looks like a medical device. Our design keeps a human making that call. We design each function to sit on the safe side of the line, and where a function cannot, we notify rather than seek approval.

Mission language versus product labelling. This document talks about keeping people safe and detecting falls, because that is the mission. Product labelling and marketing will be drafted to the wellness and decision-support line, and the safety promise attaches to the clinician service rather than to the software. Holding that discipline is a build constraint from day one.

United States: the wellness and decision-support line

The FDA reissued both governing guidance documents on 6 January 2026: General Wellness: Policy for Low Risk Devices, superseding the 2019 version, and Clinical Decision Support Software, superseding 2022.[18, 19]

What stays clear of regulation. Since the 21st Century Cures Act, software that maintains or encourages a healthy lifestyle, unrelated to diagnosing or treating disease, is excluded from the device definition. The 2026 wellness guidance goes further: non-invasive sensors estimating physiological parameters such as heart rate can qualify as general wellness products, provided claims avoid disease references and diagnostic thresholds.[18]

Where the line is crossed. The same guidance is explicit that alerting for clinical purposes, alerts prompting clinical action, and disease-specific thresholds disqualify a product from wellness status.[18] Tend's escalation would cross that line if the software made the clinical decision. Our design keeps it on the safe side in two ways. A licensed clinician interprets the signal and decides whether to act; the software only surfaces information. And the escalation logic stays transparent to that clinician. The 2026 CDS guidance keeps software outside the device definition when a professional can independently review the basis for a recommendation, and warns that black-box tools are more likely treated as devices.[19]

The honest caveat. This is a design discipline rather than a loophole. If we marketed the software as catching falls automatically, or let it escalate without a clinician, the exemption falls away. Our liability position and our regulatory position are the same discipline.

New Zealand: notification today, a new Act on the horizon

New Zealand is our beachhead, and its current regime is materially lighter than the US or EU.

There is no approval step today. Under the Medicines Act 1981 there is no pre-market approval system for medical devices. The only duty is notifying the WAND database within 30 days of taking a device to market, and notification is not an assessment of safety or performance.[20] Even if part of Tend's stack were treated as a low-risk device here, the obligation is to notify. That removes approval timelines that would otherwise gate a pilot.

The window has a clock on it. The Government is developing a Medical Products Bill to replace the Medicines Act, intended for introduction to Parliament in 2026 and taking effect around the end of the decade, with a longer transition for devices.[21] We treat the light regime as a launch window rather than a permanent state. We hold safety documentation to the incoming standard from the start, because Medsafe can request it now and will require it later.

European Union: the reason we minimise biometrics

Competitors relying on facial recognition and emotion detection carry heavy EU exposure. The official basis is worth stating, because avoiding it is a deliberate design choice for us.

Emotion recognition is high-risk by default. The EU AI Act lists AI systems intended for emotion recognition among its high-risk category, which carries strict obligations.[22] A narrow exception exists for systems designed for medical or safety purposes, which a safety-focused eldercare use may fit, though high-risk obligations still apply.[22]

Biometric data is doubly protected. Under the GDPR, biometric data used to identify a person is a special category with heightened protection, layered on top of the AI Act.[23]

Our design minimises biometric processing. We favour contactless movement and vital-sign sensing over facial recognition and emotion inference. That keeps us clear of the heaviest EU obligations our competitors invite.

The offshore clinician: an open question we flag honestly

The long-term model envisions a remote nurse, possibly offshore, triaging into many homes. This raises scope-of-practice and licensing questions that vary by country and are unsettled.

A clinician who assesses and directs care into a patient's home is generally practising in the patient's jurisdiction, which usually requires local registration. Triage and escalation may be treated differently from hands-on care, and the boundary is untested for this model.

We do not claim this is solved. The offshore ratio that drives long-run margin depends on regulatory clarity we will pursue with counsel. Appendix 2 shows the onshore fallback the model must survive if the answer is no.

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